



Size and stress dependences in the tensile stress relaxation of thin copper wires at room temperature

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ABSTRACT

The tensile stress relaxation behavior of thin copper wires with diameters of 20, 35 and 50 μm is experimentally investigated at room temperature. It is revealed that the stress relaxation can occur at low starting stress below the yield strength at room temperature. Significant size and stress dependences in the stress relaxation are observed. That is at a given stress, a smaller-diameter sample exhibits a much faster stress relaxation. At a given diameter, the stress relaxation is accelerated with increasing the starting stress. The value of activation volume increases with increasing the wire diameter, while it increases with decreasing the starting stress. The strain rate elevates as the wire diameter decreases or the starting stress increases. A bilinear relation between the logarithmic strain rate and the logarithmic stress during the relaxation is observed only for the wire sample with a diameter of 20 μm . The intersection of dislocations with grain boundaries is considered to be the main deformation mechanism during the stress relaxation. A rapid exhaustion of mobile dislocations is observed in the wire with smaller diameter in the primary of relaxation. It is attributed to annihilation of dislocations easily escaping from the surface due to larger fraction volume of the near-surface zone. After the primary regime, the higher possibility of dislocations being pinned or immobilized by grain boundaries in a larger-diameter sample is accounted for the size dependence.

1. Introduction

It is widely accepted that the mechanical properties of metals at small scales can be affected by grain size or specimen size, exhibiting size dependence. For the influence of grain size on small scale plasticity, the famous Hall-Petch (H–P) relation (Hall, 1951; Petch, 1953), illustrates the effect of grain size d on the yield strength σ_Y , giving $\sigma_Y \propto d^{-1/2}$ (Armstrong, 2014). The specimen size effect at small scales has also been experimentally investigated in various loading conditions, including wire torsion (Dunstan et al., 2009; Fleck et al., 1994; Gan et al., 2014; Guo et al., 2017a; Liu et al., 2012a, 2013a; 2013b), foil bending (Ehrler et al., 2008; Haque and Saif, 2003; Hayashi et al., 2011; Moreau et al., 2005; Stölken and Evans, 1998), micro- and nano-indentation (Ma et al., 2012; Swadener et al., 2002; Zhu et al., 2008), and tension or compression of micro- and nano-pillars (Greer and De Hosson, 2011; Jennings et al., 2010; Kaufmann et al., 2011; Kiener et al., 2010; Kim et al., 2012a; Soler et al., 2012; Uchic et al., 2004). Besides, the size effect at the onset of plasticity is certainly observed in several experiments (Dunstan et al., 2009; Greer and De Hosson, 2011; Guo et al., 2017a; Liu and Dunstan, 2017; Zhu et al., 2008). Not merely the specimen size effect in small scale plasticity, an elevated bending stiffness in a nickel foil with decreasing thickness is recently observed by Lei et al. (2016) in the elasticity.

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Understanding the time-dependent plasticity is important in engineering applications. Both creep and stress relaxation behaviors of metallic materials play important roles in designing mechanical component. It has been reported, for example, Armstrong (2017, 2006), Conrad (2004) and Rodriguez (2004), that the grain size has an impact on the time-dependent plasticity for metallic materials. Typically, an analogous Hall-Petch relation between the activation volume V and the grain size d has been proposed (Armstrong and Balasubramanian, 2017; Armstrong and Rodriguez, 2006; Rodriguez, 2004), i.e., $1/V \propto d^{-1/2}$. Conrad (2004) concluded that the rate-controlling mechanism for the regime $d > \sim 1 \mu\text{m}$ is the intersection of dislocations. When the grain size ranges from $d \approx 10 \text{ nm}$ to $1 \mu\text{m}$, the mechanism is proposed to be the grain-boundary shear promoted by the pileup of dislocations. For the regime, it appears to be the grain-boundary shear by the applied stress. Further investigations on creep or stress relaxation behaviors of samples with size or twin thickness at a few microns and sub-microns show that the time-dependent plasticity depends on specimen size or twin thickness (Choi et al., 2013; Gu and Ngan, 2014; Guisbiers and Buchaillot, 2008; Lu et al., 2009a; Ma et al., 2008; Shen et al., 2006; Yang et al., 2016a, 2016b). For instance, a pillar diameter-dependent creep behavior of nanocrystalline (nc) Ni was observed by Choi et al. (2013). It is revealed that a higher total strain and strain rate accrue in the pillar with a smaller diameter. Lu et al. (2009) found an increase of activation volume with decreasing twin thickness of nano-twinned (nt) copper. A value of about $20b^3$ and $36b^3$ (b is the Burger's vector) was derived for 15 nm and 35 nm twin thicknesses nt samples, respectively. Shen et al. (2006) investigated the strain rate sensitivity of ultrafine-crystalline copper with different concentrations of nanoscale growth twins. A higher twin density leading to an enhanced strain rate sensitivity was indicated. According to the observations, the author found that the time-dependent deformation will be accelerated with decreasing sample size or twin thickness at sub-microns.

At a few microns, Gu and Ngan (2014) studied the creep behavior of duralumin micro-pillars with diameters ranging from $\sim 1 \mu\text{m}$ to $\sim 6.5 \mu\text{m}$. An enhanced creep strain rate in a larger pillar is observed. This phenomenon contrasts to aforementioned observations, where a rapid time-dependent deformation occurs in a sample with a smaller specimen size. It was explained that the dislocations are not easily pinned in this range. As the pillar size increases, the dislocations take longer time to move across the specimen. Hence the retained dislocation density is much higher, which leads to a higher strain rate. Nevertheless, they predicted that the trend of creep rate with specimen size would be reversed for larger specimens, probably in tens of microns. In this case, dislocations experience a higher chance of being pinned and immobilized by the obstacles.

In order to study the time-dependent plasticity of samples with size in the tens of microns, we experimentally explore the tensile stress relaxation behavior of thin copper wires with different diameters at room temperature.

2. Experimental details

The commercially cold drawn polycrystalline copper wires (99.999% purity) with diameters of 20, 35, 50 μm were employed. All as-received wires were annealed at 410 °C for 2.5 h in a vacuum furnace before tests. The annealing aims to reduce defects and ensures different-diameter samples possessing similar grain size. To figure out the microstructures of undeformed wires, both cross- and longitudinal-sections FIB images were given by using a Quanta 3D Dual Beam FIB system. The microstructure texture and average grain size were determined by the electron back scattered diffraction (EBSD) recorded in an SEM (Carl Zeiss, GeminiSEM300, Germany). Then the material characteristic was figured out by conducting tensile tests using a micro-tensile tester (Liu et al., 2016, 2012b). The corresponding experimental apparatus is given in Fig. 2(a). In the tension tester, the data acquiring and operating system was designed based on LabVIEW. The load cell of the tester consists of a noncontact type laser displacement sensor and a thin beam with thickness of 0.45 mm fixed at both ends. The laser displacement sensor was used to detect the deflection of the beam. And then the load on the specimen is determined by calibration experiment. The load cell of the tester has a measurement accuracy of 0.1 mN, with maximum measurement range of about 1 N. Before mounting, the wire was glued both ends in a prepared sample card with epoxy resin. Then the sample card was clamped to the top and bottom grips. The bottom grip connects to a high precision linear motor (PI M227.25). Cutting both sides of the sample card, the deformation of sample is recorded as the displacement of linear motor subtracting that of the thin beam in the tension. A detailed description of the experimental setup and data evaluation can be found in Liu et al. (Liu et al., 2012a, 2012b). The sample card has a gauge length around 10 mm here for all tests. The experimental data were handled with: the engineering stress $\sigma = F/A$ and the engineering strain $\epsilon = \Delta l/l$, where F denotes the applied force, A the area of cross-section, Δl the stretched length and l the initial length (gauge length).

For the tensile stress relaxation tests, the aforementioned tension tester was also employed. During the tests, samples with the same gauge length of 10 mm were stretched at a strain rate of $10^{-3}/\text{s}$ to the prescribed starting stress. Then the tensile driver was maintained to allow stress relaxation. The decay stress was recorded as a function of holding time. For repeated transients, the relaxed specimen was reloaded to the same starting stress. Here, single stress decay tests were conducted at starting stress of 62, 77, 92 MPa with holding time of about 1500 s. In order to figure out the evolution of mobile dislocations during the stress relaxation, repeated relaxation tests were performed with a starting stress of 120 MPa. In each relaxation cycle, a holding time of about 200 s was kept. All tests were conducted at room temperature. At least three specimens were carried out for each diameter sample to obtain a reproducible result and to decrease the random error.

3. Results

3.1. Fundamental characteristic

The cross-section and longitudinal-section FIB images for undeformed wires with different diameters are given in Fig. 1(a)–(f). One can see that grains in the cross- and longitudinal-sections keep similar for wires with different diameters. The grain elongation

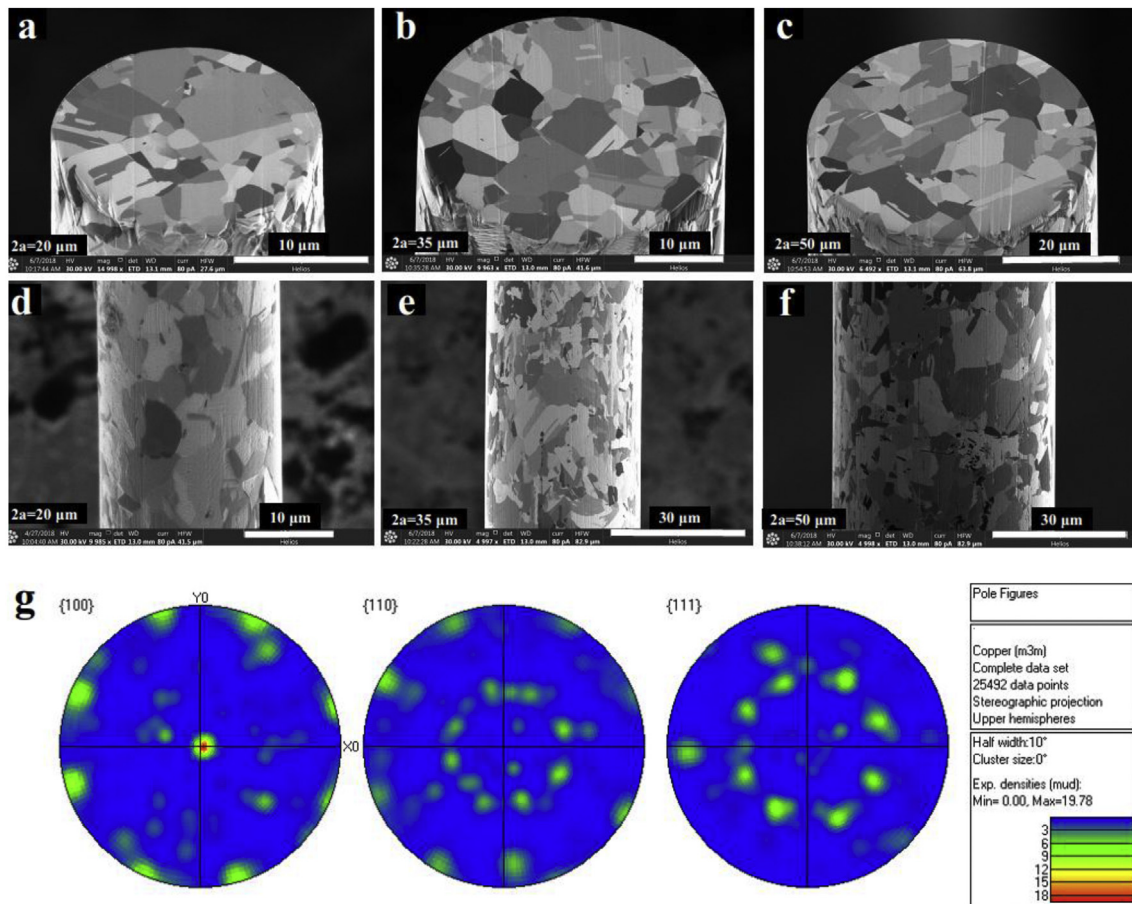


Fig. 1. The microstructures of the undeformed copper wires. (a)–(f) Typical cross- and longitudinal-section FIB images for wires with different diameters. (g) The EBSD pole figure illustrating the texture in the polycrystalline copper wires. Where “a” denotes the radius.

along the drawing direction is not obvious after the annealing process. The corresponding pole figure for undeformed wires is shown in Fig. 1(g). The wires are found to be substantially isotropic. The random texture illustrated in the EBSD pole figure confirms that grain orientation has little effect on the mechanical behavior of the wires. The average grain size, derived from grain distributions in EBSD grain orientation maps, for wires with different diameters possesses a similar value of about 4.2 μm. Thus the influence of grain size on time-dependent plasticity can be neglected for our tests. The engineering stress-strain curves for the wires are illustrated in Fig. 2(b). One can see that all stress-strain curves almost keep coincident. The yield strength for wires with different diameters has an identical value of about 110 MPa. This agrees with the previous findings (Guo et al., 2017a). That is the tensile properties at tens of micron meters would not be affected by the specimen size. Reminding of the aforementioned chosen starting stresses, one can see that the stress relaxation tests were conducted both in the elastic and plastic regions.

3.2. Stress relaxation results

For the stress relaxation under tensile condition, the effect of possible misalignment can be neglected comparing with that in the compressive creep of micro-pillars. Compared with the complicated stress condition in the indentation creep, there is a simple and uniform stress-distribution over the cross-section and along the axis of the wire under tension. Typical engineering stress-strain response, including load process and stress-relaxation process, for sample with diameter of 35 μm is depicted in Fig. 3(a). One can see that the stress relaxation does occur in elastic region (below the yield strength) at room temperature. Stress relaxation observed at low stress in nc metals confirms that plasticity happens below the yield strength, which is attributed to their very high fraction of grain boundaries acting as strong sources of dislocations (Gianola et al., 2006). However, our findings indicate that plasticity could also occur below the yield strength without high fraction of grain boundaries since only several grains exist in the cross-section of the wires. For instance, there are only about 6 grains in the cross-section of wire with a diameter of 20 μm. This phenomenon agrees with the previous observation on stress relaxation test under non-uniform deformation (Guo et al., 2017b). Typical results of engineering stress versus holding time for Cu samples with diameters of 20, 35, 50 μm at a starting stress of 77 MPa are described in Fig. 3(b). A usual transient regime at the primary stage of relaxation (about $t \approx 300$ s) can be seen. After the transient regime, the stress drop

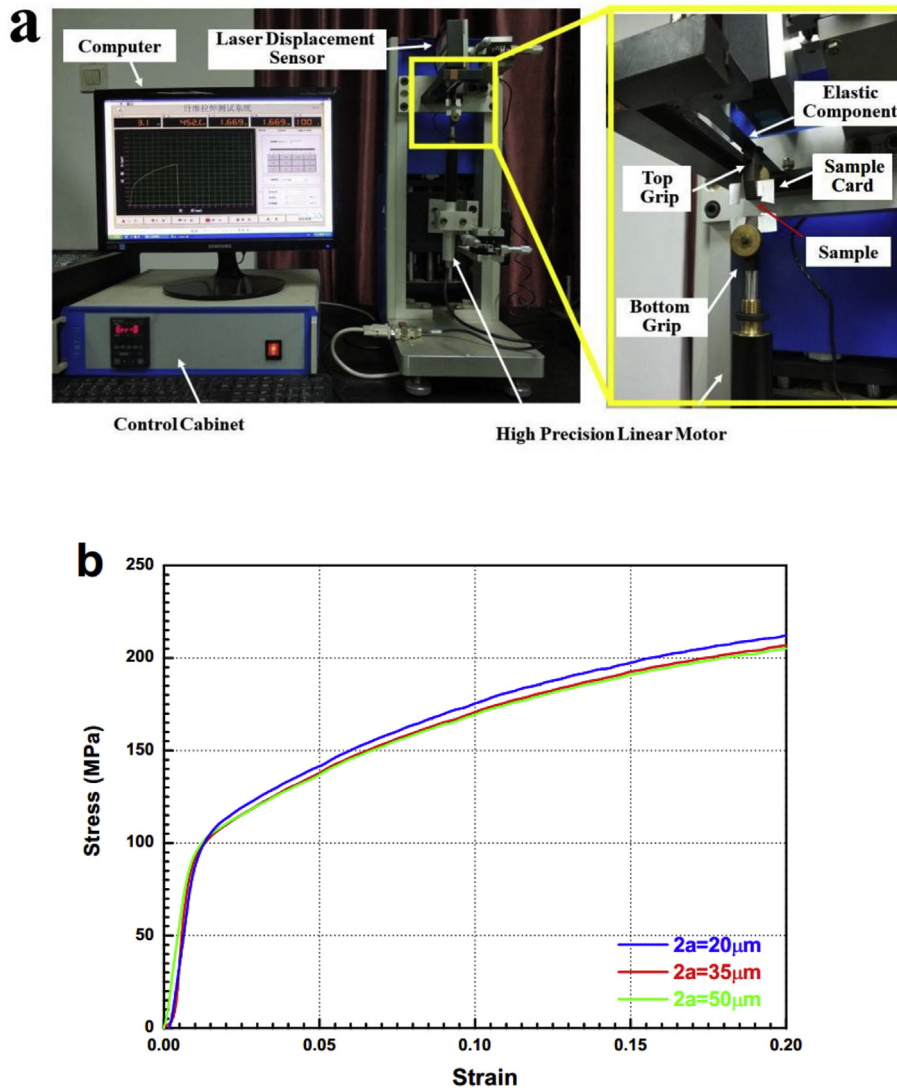


Fig. 2. (a) Experimental apparatus of micro-tensile tester. (b) Fundamental tensile engineering stress-strain responses for samples with diameters of 20, 35, 50 μm , respectively.

decreases with time much slowly.

According to the data in Fig. 3(b), one can obtain the results of decay stress versus holding time for samples with different diameters, as shown in Fig. 3(c). The stress relaxation displays a significant size effect whereby the smaller the sample diameter, the rapider the stress relaxation. For instance, at $t \approx 1500$ s, the total decay stress is about 36, 24, 15 MPa for the samples with diameters of 20, 35, 50 μm , respectively. The finding here is in agreement with the previous reports on nc nickel (Choi et al., 2013) or nt copper (Lu et al., 2009b). Nevertheless, the phenomenon contrasts to that observed by Gu and Ngan (2014), where a rapider creep in a larger sample was observed. Fig. 3(d) illustrates the evolution of decay stress with holding time for the Cu sample with a diameter of 20 μm at various starting stresses. A stress dependence is revealed, reflecting that the decay stress increases with increasing the starting stress. For example, at the end of the stress relaxation, the total decay stress is about 18, 24, 30 MPa for the wire at a starting stress of 62, 77, 92 MPa, respectively. It is consistent with that observed in nc Ni (Mohanty et al., 2016), duralumin micro-pillars (Gu and Ngan, 2014), nc Ni pillars (Choi et al., 2013) and thin copper wires (Guo et al., 2017b). The results of decay stresses in single transient versus sample diameters at $t \approx 1500$ s for starting stress of 62, 77, 92 MPa are shown in Fig. 4. It is evident that, at a given stress, rapider stress relaxation occurs in a sample with a smaller diameter. In addition, at a given diameter, the stress relaxation is accelerated with increasing the starting stress.

The stress decay versus holding time results during the three relaxation cycles for the samples with diameters of 20, 35, 50 μm at a starting stress of 120 MPa are given in Fig. 5. One can see that the decay stress slows down along the series for each sample. For instance, the decay stress is about 32, 26, 22 MPa in the 1st, 2nd, 3rd relaxation cycle for the wire with a diameter of 20 μm . The slowing down may arise as a result of the hardening subsequent to plastic deformation and to decrease in mobile dislocation density

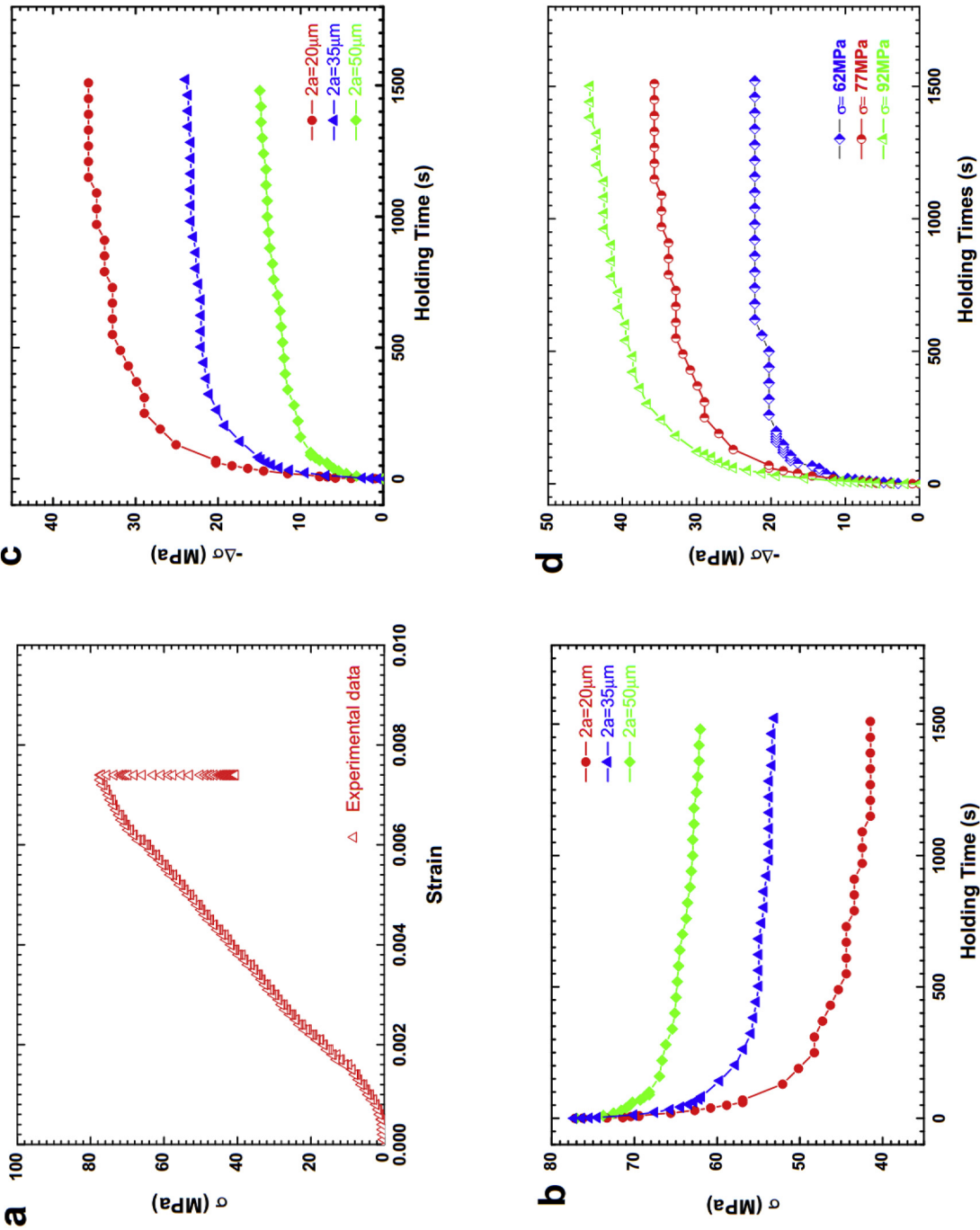


Fig. 3. (a) Typical stress-strain curve, including load process and stress relaxation process, for wire with a diameter of 35 μm at a starting stress of 77 MPa. (b) and (c) show the typical stress decay stress versus holding time for samples with diameters of 20, 35, 50 μm Cu wires at a starting stress of 77 MPa. (d) Evolution of the decay stress with holding time for the wire with a diameter of 20 μm at different starting stresses.

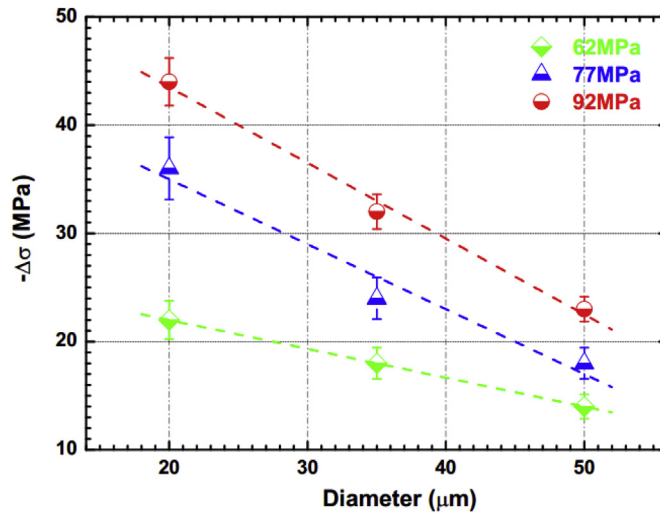


Fig. 4. Total decay stress, with $t \approx 1500$ s, versus sample diameters at starting stress of 62, 77, 92 MPa, respectively.

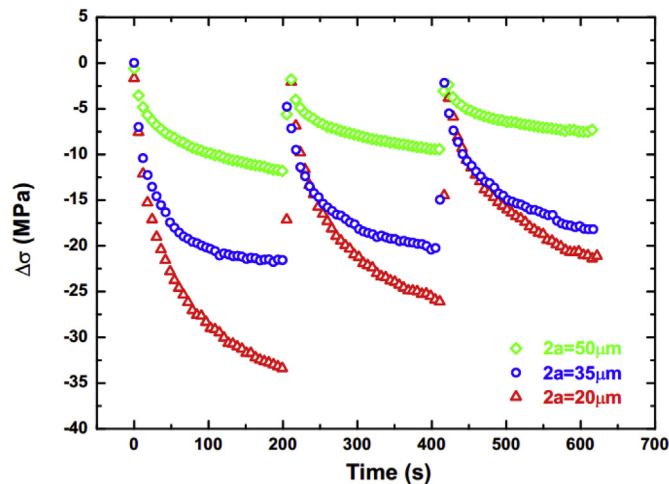


Fig. 5. Typically repeat stress relaxation series for samples with diameters of 20, 35, 50 μm at a starting stress of 120 MPa.

along the series (Caillard and Martin, 2003). A size dependence in each relaxation cycle is also observed. That is the smaller the wire diameter, the rapider the decay stress.

3.3. Activation parameters

The decrease of stress during relaxation is considered by an increase in plastic strain ε_p which relieves the elastic strain σ/M (Caillard and Martin, 2003). Here, M represents the nominal Young's modulus of materials including the machine influence (determined as 107 GPa in current study). The relation between the total strain ε_t and the plastic strain ε_p during the stress relaxation is:

$$\varepsilon_t = \varepsilon_p + \frac{\sigma}{M} \quad (1)$$

Considering the stress relaxation condition where the total strain is fixed, i.e., $\dot{\varepsilon}_t = 0$, thus one can obtain

$$\dot{\varepsilon}_t = \dot{\varepsilon}_p + \frac{\dot{\sigma}}{M} = 0 \quad (2)$$

where $\dot{\varepsilon}_t = 0$ denotes the total strain rate, $\dot{\varepsilon}_p$ the relaxation strain rate and $\dot{\sigma} = 0$ the relaxation stress rate, respectively.

Generally, in a single relaxation, the decay stress is often expressed in the form of shear stress $\Delta\tau$ and the logarithmic function of holding time t . That is

$$\Delta\tau(t) = -\frac{kT}{V_a} \ln\left(1 + \frac{t}{c_r}\right) \quad (3)$$

where k represents the Boltzmann constant, T the absolute temperature, V_a the apparent activation volume of the thermally activated process, and c_r the corresponding time constant. The magnitude of the activation volume V_a is expressed as $\sim b^3$. Here, b is the Burger's vector, $b = 0.245$ nm for Cu.

Note that the activation volume is defined in terms of the shear stress, while the stress relaxation test is conducted under tensile condition. Here, the shear stress and the normal stress can be related by $\sqrt{3}\tau = \sigma$ by introducing the von Mises yield criterion (Caillard and Martin, 2003). According to Eq. (3), the decay stress $\Delta\sigma$ in the tensile stress relaxation can be expressed by

$$\Delta\sigma(t) = -\frac{\sqrt{3}kT}{V_a} \ln\left(1 + \frac{t}{c_r}\right) \quad (4)$$

Combining Eq. (2) with Eq. (4), one can obtain the plastic strain rate as

$$\dot{\epsilon}_p = \frac{\sqrt{3}kT}{MV_a} \left(\frac{1}{t + c_r}\right) \quad (5)$$

The strain rate sensitivity parameter m , reflecting the strain dependence on stress, is defined by

$$m = \frac{\partial \log \sigma}{\partial \log \dot{\epsilon}_p} \bigg|_T = \frac{\partial \ln \sigma}{\partial \ln \dot{\epsilon}_p} \bigg|_T \quad (6)$$

The physical activation volume V^* in repeated relaxation tests, characterizing the stress sensitivity of dislocation velocity, can be determined by

$$V^* = \sqrt{3}kT \frac{\ln(\dot{\epsilon}_{i2}/\dot{\epsilon}_{f1})}{\Delta\sigma_{i2-f1}} \quad (7)$$

where $\dot{\epsilon}_{i2}$ and $\dot{\epsilon}_{f1}$ denote the strain rate at the onset of next relaxation and that at the end of first relaxation, respectively. $\Delta\sigma_{i2-f1}$ represents the nominally elastic loading between two relaxation segments.

The dislocation velocity, in a short transient, has the form

$$v = v_0 \exp\left(\frac{V^* \Delta\tau^*}{kT}\right) \quad (8)$$

where v_0 is the dislocation velocity at the onset of the transient, and $\Delta\tau^*$ is the change of stress. The mobile dislocation density ρ_m , reflecting the effect of the structural evolution of dislocation on plastic deformation (Lu et al., 2009a), can be expressed by a power relation in terms of dislocation velocity as

$$\frac{\rho_m}{\rho_{m0}} = \left(\frac{v}{v_0}\right)^\beta \quad (9)$$

where ρ_{m0} is the initial mobile dislocation density. It follows by combining Eqs. (7)–(9) that

$$\frac{\rho_m}{\rho_{m0}} = \left(\frac{c_r}{c_r + t}\right)^{\frac{\beta}{1+\beta}} \quad (10)$$

where β denotes a dimensionless immobilization parameter, which is calculated by

$$\beta = \frac{\Omega}{1 + K/M} - 1 \quad (11)$$

where $\Omega = V_a/V^*$ and K is the strain-hardening coefficient, calculated by $K = \dot{\sigma}/\dot{\epsilon}_p$.

Based on Eq. (4), the fitted results of apparent activation volume for samples at a starting stress of 77 MPa is given in Fig. 6(a). The prediction gives the values of apparent activation volume as 95.3 b^3 , 126.8 b^3 , 160.6 b^3 for wires with diameters of 20, 35, 50 μm , respectively. A size effect in the apparent activation volume is revealed. Namely the apparent activation volume increases with increasing the wire diameter. The fitting results of the activation volumes for the sample with a diameter of 35 μm at different starting stresses is illustrated in Fig. 5(b). One can see the activation volume increases with decreasing the starting stress. For instance, the value is about 164.3 b^3 , 126.8 b^3 , 99.4 b^3 at starting stress of 62, 77, 92 MPa, respectively. The fitted values of apparent activation volumes for wires with different diameters at different starting stresses in single transient are demonstrated in Table 1. To sum up, the apparent activation volume increases with increasing the wire diameter. While it increases with decreasing the starting stress. The estimated values of apparent and physical activation volumes versus wires with different diameters at a starting stress of 120 MPa are given in Table 2. One can see that the value of physical activation volume is less than that of apparent activation volume in each relaxation cycle. And the apparent activation volume decreases with increasing relaxation cycle. The physical activation volume increases with increasing relaxation cycle. Our findings keep consistent with those observed in Lu et al. (2009) and Wang et al. (2006). E.g., for the wire with a diameter of 20 μm , the apparent activation volume has a value of about 68.6 b^3 and 64.3 b^3 , the

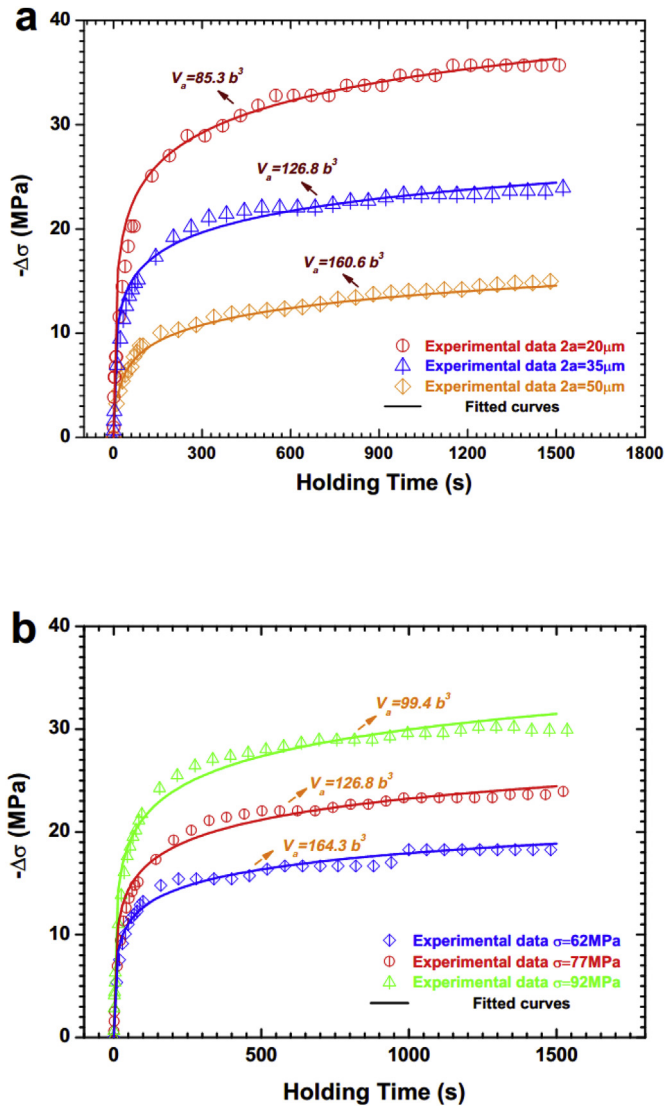


Fig. 6. Experimental data and fitted curves versus holding time (a) for samples with diameters of 20, 35, 50 μm at a starting stress of 77 MPa, (b) for wire with a diameter of 35 μm at different starting stresses.

Table 1

The value of apparent activation volume V_a and strain rate sensitivity parameter m versus wires with different diameters and at different starting stress in current study.

Diameter	Starting stress					
	62 MPa		77 MPa		92 MPa	
	V_a	m	V_a	m	V_a	m
20 μm	120.5 b^3	0.034 1st 0.078 2nd	95.3 b^3	0.036 1st 0.094 2nd	78.9 b^3	0.039 1st 0.108 2nd
35 μm	164.3 b^3	0.047	126.8 b^3	0.054	99.4 b^3	0.061
50 μm	191.7 b^3	0.033	160.6 b^3	0.036	135.2 b^3	0.037

physical activation volume is about 47.6 b^3 and 51.6 b^3 in the 1st and the 2nd relaxation cycle, respectively. A size dependence in physical activation volume is also observed. The value of physical activation volume increases with the increasing wire diameter in each relaxation cycle. For example, the physical activation volume is about 88.1 b^3 , 61.4 b^3 , 47.6 b^3 for the wire with diameters of 50, 35, 20 μm in the 1st relaxation cycle. The current value of activation volume is much smaller than that of the conventional coarse-

Table 2The value of apparent activation volume V_a and physical activation volume V^* versus wires with different diameters at a starting stress of 120 MPa.

Diameter	1 st		2 nd		3 rd	
	V_a	V^*	V_a	V^*	V_a	V^*
50 μm	112.7 b ³	88.1 b ³	107.8 b ³	91.9 b ³	104.5 b ³	–
35 μm	85.2 b ³	61.4 b ³	82.1 b ³	64.5 b ³	75.0 b ³	–
20 μm	68.6 b ³	47.6 b ³	64.3 b ³	51.6 b ³	59.5 b ³	–

grained (cg) Cu (about 600 b³) (Yang et al., 2016a). It is also smaller than that of ultrafine-grained (ufg) Cu (about 350 b³) (Lu et al., 2009b). But, the value is slightly larger than that of nt Cu (about 20 b³) (Lu et al., 2009b).

The logarithmic form of the strain rate versus holding time at a starting stress of 92 MPa for wires with diameters of 20, 35, 50 μm in single transient is shown in Fig. 7. One can see that the strain rate drops quickly with holding time at the primary stage of the stress relaxation. After the rapid-drop stage, the strain rate changes slowly with holding time. From Fig. 7, one can obtain that the strain rate is in the range from $\sim 10^{-8} \text{ s}^{-1}$ to $\sim 10^{-4} \text{ s}^{-1}$. The strain rate is comparable with that observed in cg metal (Yang et al., 2016a). It is slightly smaller than those of ng and nt metals (ranging from 10^{-8} s^{-1} – 10^{-4} s^{-1}) (Yang et al., 2016a). A size effect in the strain rate is revealed. That is at a given holding time, the wire with a diameter of 20 μm has the highest strain rate. While the wire of 50 μm possesses the lowest one. Here, a quasi-steady strain rate is estimated as the average value of the strain rate for the holding times between 1400 s and 1500 s in single transient. The quasi-steady strain rate versus wire diameters at different starting stresses in single transient is given in Fig. 8. A stress dependent feature can also be obtained. The strain rate increases with decreasing the wire diameter, and increases with increasing the starting stress. For example, at a starting stress of 92 MPa, the quasi-steady strain rate decreases from $\sim 3.3 \times 10^{-8} \text{ s}^{-1}$ to $\sim 1.7 \times 10^{-8} \text{ s}^{-1}$ as the wire diameter increases from 20 μm to 50 μm . For the wire with a diameter of 20 μm , the strain rate increases from $\sim 1.9 \times 10^{-8} \text{ s}^{-1}$ to $\sim 3.3 \times 10^{-8} \text{ s}^{-1}$ as the starting stress increases from 62 MPa to 92 MPa.

The logarithmic plot of strain rate against the stress during the stress relaxation at a starting stress of 77 MPa for wires with different diameters is given in Fig. 9(a). One can see that an almost linear relation between the strain rate and the stress is observed for the wires with diameters of 50 μm and 35 μm . Surprisingly, a bilinear relation is found for wire with a diameter of 20 μm . The first (1st) linear stage occurs in the initial stage of the stress relaxation. After that, the second (2nd) linear stage happens. The logarithmic plots of plastic strain rate and stress during the relaxation for the wire with a diameter of 20 μm are given in Fig. 9(b). A bilinear relation can be observed at different starting stresses. We find that the transition stress is about 54 MPa, 67 MPa, 81 MPa with starting stress of 62 MPa, 77 MPa, 92 MPa, respectively. Associated with the decay stress results as shown in Fig. 3 (d), this indicates that the first linear stage occurs within $t \approx 20$ s. In this range, the mobile dislocations exhaust with time very quickly (see Fig. 10). Yang et al. (2016a) proposed a three-stage relaxation process, from a linear high stress region through a subsequent nonlinear stress region and finally to a linear low stress relaxation region, for cg, ng and nt Cu specimens at temperature higher than 22 °C, 22 °C, 30 °C, respectively. Our study shows that the bilinear relaxation can occur at room temperature, but it is just observed for wire with a diameter of 20 μm . In contrast to that in Yang et al. (2016a), the subsequent nonlinear stress region is unobvious. According to Eq. (6), two strain rate sensitivity parameters will be derived corresponding to the 1st and 2nd linear stages by linear fitting. The values of the strain rate sensitivity parameter for the wires with different diameters and at different starting stresses are summarized in Table 1. The fitting gives the value ranging from 0.033 to 0.108, which is comparable with that of ng and nt metals (about 0.02–0.1) (Chen et al., 2006; Lu et al., 2001, 2005; Yang et al., 2016b), and slightly larger than that of ufg metals (about 0.01–0.03) (Dalla Torre

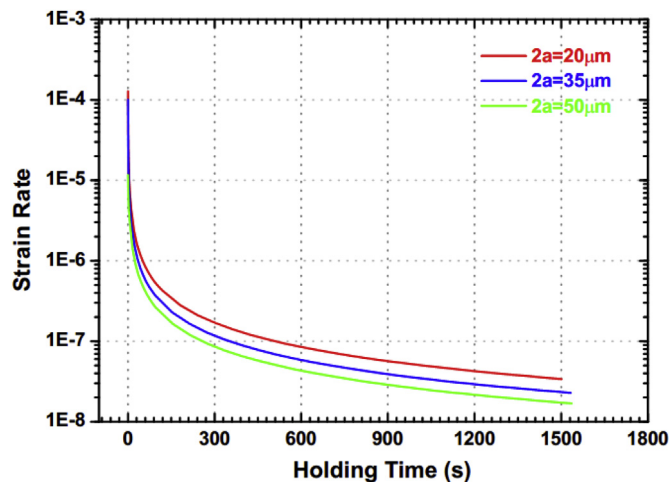


Fig. 7. The logarithmic form of strain rate versus holding time during the stress relaxation with a starting stress of 92 MPa in single transient for wires with diameters of 20, 35, 50 μm , respectively.

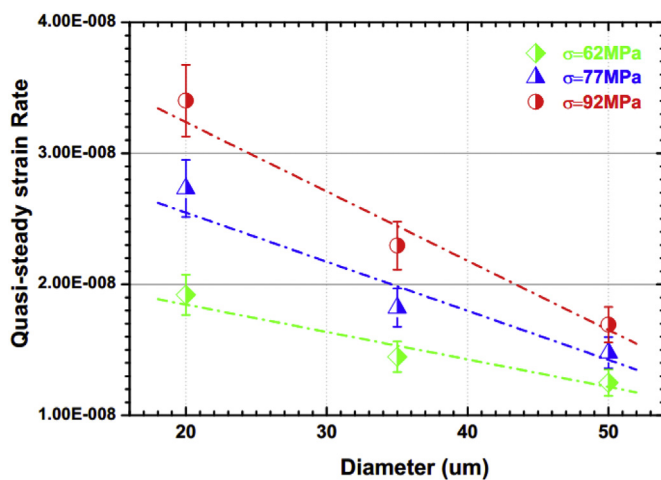


Fig. 8. Quasi-steady strain rate versus wire diameters at starting stresses of 62, 77, 92 MPa.

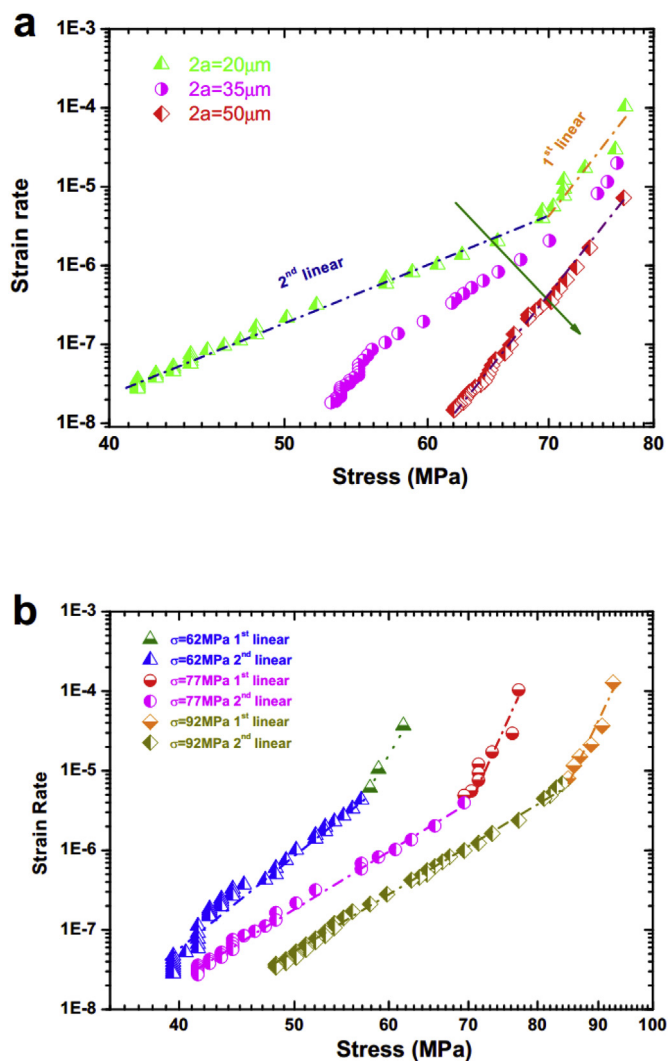


Fig. 9. The logarithmic strain rate versus logarithmic stress during relaxation (a) for wires with different diameters at a starting stress of 77 MPa, (b) for wire with a diameter of $20 \mu\text{m}$ at different starting stresses.

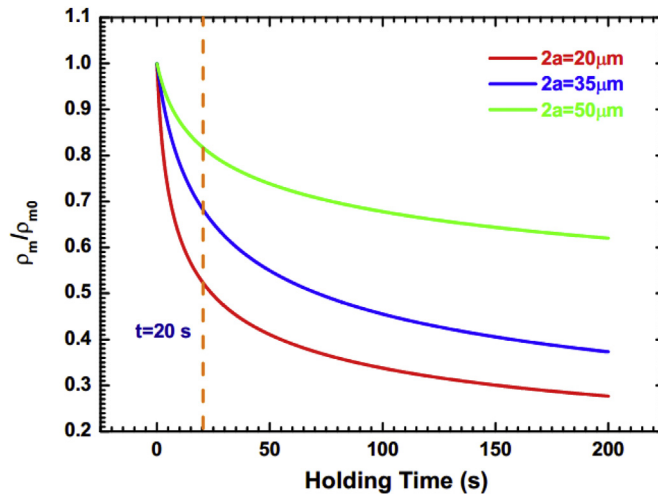


Fig. 10. Evolution of the mobile dislocation density for wires with diameters of 20, 35 and 50 μm in the 1st relaxation cycle at a starting stress of 120 MPa.

et al., 2002; Zhu et al., 2007). But the value is obviously different from that of cg counterparts (about 0.002–0.007) (Follansbee and Kocks, 1988; Follansbee et al., 1984). The value is also different in the 1st and the 2nd linear stage for the wire with a diameter of 20 μm . The strain rate sensitivity parameter in the 1st linear stage is much less than that in the 2nd linear stage. From Table 1, a significant size effect is demonstrated. The strain rate sensitivity parameter increases with decreasing the wire diameter. For instance, at a starting stress of 62 MPa, the value is 0.033, 0.047, 0.078 (the 2nd linear stage) for the wire with a diameter of 50, 35, 20 μm , respectively. Meanwhile, a stress dependence is also reflected. The strain rate sensitivity parameter increases with increasing the starting stress. The positive value of m indicates that the strain rate increases with increasing the starting stress (Yang et al., 2016b), which is consistent with our results. For example, the value of m for the wire with a diameter of 20 μm increases from 0.047 to 0.061 with increasing the starting stress from 62 MPa to 92 MPa.

Fig. 10 gives the evolution of mobile dislocations for wires with diameters of 20, 35, 50 μm in the 1st relaxation cycle at a starting stress of 120 MPa. One can see that the mobile dislocation density decreases as the holding time increases. And a size effect is observed. That is the smaller of diameter, the larger decrease of ρ_m/ρ_{m0} . For instance, at the end of 1st relaxation cycle, ρ_m/ρ_{m0} decreases about 0.72, 0.63, 0.38 for wires with diameters of 20, 35, 50 μm , respectively. The large reduction of ρ_m/ρ_{m0} indicate that few mobile dislocation density remain after stress relaxation. The value $\Delta\rho_m/\rho_{m0}$ observed here is evidently larger than that of about 0.15–0.22 for nt Cu, comparable with of about 0.36 for ufg Cu observed in Lu et al. (2009).

4. Discussion

4.1. Deformation mechanism

The activation parameters, e.g., the activation volume and the strain rate sensitivity parameter, could shed light on the deformation mechanism during the stress relaxation. It is generally accepted that for cg Face-Centered Cubic (FCC) metals, a rate determined process, such as the intersection of forest dislocations, possesses a large value of the activation volume at the magnitude of several hundreds of b^3 and above. At another extreme, grain boundary diffusive mediated creep (Coble creep) or lattice diffusion mediated creep (Nabarro-Herring creep) may give a small value of the activation volume, e.g., less than $1 b^3$. If the value of the activation volume ranges from several b^3 to $100 b^3$, the rate process typically involves dislocation cross-slip or dislocation intersection with grain boundary. Coble (1963) and Lüthy et al. (1979) concluded that the dislocation-mediated plasticity corresponds to a low strain rate sensitivity parameter ($m < 0.1$). If the strain rate sensitivity parameter is around unity ($m \sim 1.0$), it denotes the grain boundary diffusion creep. If the value of m ranges from 0.1 to 1, the mechanism might be related to the grain boundary sliding or/and grain rotation.

In this study, since the value of the activation volume ranges from $68.9 b^3$ – $188.7 b^3$ and the strain rate sensitivity parameter is in the range 0.032–0.108, the mechanisms of grain boundary diffusive mediated creep and lattice diffusion mediated creep are firstly excluded. The intersection of forest dislocations can also be eliminated, since a less value of the activation volume is observed in our tests. Hence the dislocation cross-slip or the dislocation intersection with grain boundary may dominate the deformation mechanism. Connecting the deformation mechanism with the observed size dependence is essential. When a mobile dislocation glides in a crystal, it may encounter precipitate particles or grain boundaries. Upon interacting with such a particle or grain boundary, the dislocation may either overcome them by some bypass mechanisms such as cross-slip (Poirier, 1985) or may be pinned and immobilized by it (Shen et al., 1988). In Gu and Ngan (2014), the dislocation is considered not to be easily pinned or immobilized by the precipitates or grain boundaries with the samples size at a few microns. When a dislocation encounters them, it overcomes them by the bypass

mechanism. In a larger specimen, a dislocation spends longer time to travel across the specimen. Hence, the retained dislocation density is higher, resulting in a higher strain rate. This contrasts to our findings where larger strain rate occurs in a smaller specimens. So, the dislocation cross-slip should not be the main deformation mechanism in this study.

We propose that intersection between dislocations and grain boundaries is the main deformation mechanism. Grain boundaries, acting as dislocation obstacles and sources (Brentnall and Rostoker, 1965; Kurzydowski et al., 1984; Shen et al., 1988; Worthington and Smith, 1964), play important role in the motion of dislocations. There are several mechanisms for slip propagation by the movement of dislocations across grain boundaries, including dislocation nucleation in grains, dislocation absorption by dissociation in the interface, dislocation transmission across the interface and dislocation absorption and subsequent re-emission (Shen et al., 1988). For each of these mechanisms, dislocation generation, destruction, and motion may be required in the grain boundary. An evidence indicates that a high density of dislocations during plastic flow is present around the grain boundaries (Hirth, 1972). The controlling deformation process can be mediated by dislocations inside the grain boundary or in a narrow grain boundary affected zone (Kumar et al., 2003). A dislocation can be pushed to punch through the obstacle of a dense “forest” of dislocations at the grain boundary, which is analogous to that of the dislocation intersection mechanism in normal FCC metals (Caillard and Martin, 2003). The difference is that the “forest” dislocations providing the short-range resistive force against dislocation motion are now localized in bundles and concentrated to small spacing at the grain boundary, leading to a small activation volume (Wang et al., 2006).

Dislocation transfer can be delayed due to the trapped dislocations at the grain boundary (Kacher and Robertson, 2014; Polcarová et al., 2006). The thermally activated pinning-depinning mechanism indicates that not only nucleation but also propagation has a thermal component (Van Swygenhoven, 2008). When a propagating dislocation travels across the grain to the adjacent grain boundaries, two ends of the dislocation may move along the grain boundary. When the pinning occurs, which is especially likely at low temperatures, the de-pinning is thermally activated. It may be the rate-limiting step in controlling the plastic flow (Wang et al., 2006). Molecular dynamics (MD) simulation (Van Swygenhoven et al., 2006) suggests that the propagation of a dislocation would be hindered by the presence of grain boundary ledges, acting as pinning points which have to be overcome by the thermal activation. Wang et al. (2006) pointed out that the interaction between dislocations with grain boundaries increases with the decrease of grain size, due to the increase of the population of grain boundaries and sub-boundaries. For the strain rate sensitivity, many experiments have revealed that the value of strain rate sensitivity $m = 0.01$ – 0.03 in fcc metals (Dalla Torre et al., 2002; Zhu et al., 2007), and $m = 0.02$ – 0.1 in bcc and hcp metals (Chen et al., 2006; Lu et al., 2001, 2005; Yang et al., 2016b). It is obviously different from that of $m = 0.002$ – 0.007 in fcc counterparts (Follansbee and Kocks, 1988; Follansbee et al., 1984). Associated with the $m = 0.032$ – 0.102 observed here, our results agree with that an increase in m has been common for FCC metals with decrease in the characteristic of grain size or twin spacing as concluded in Yang et al. (2016b). Furthermore, an analogous Hall-Petch relation between the activation volume parameter and the grain size has been proposed by Armstrong and Rodriguez (2006). It illustrates the activation volume decreases with decreasing grain size (see Eq. (20)). The boundary-controlled dislocation activities may contribute to the elevation of strain rate sensitivity and reduced activation volume for metals with smaller grain size or twin spacing (Blum and Zeng, 2009; Duhamel et al., 2010; Wang et al., 2006; Zhou et al., 2014).

4.2. Size effect in strain rate

Gu and Ngan (2014) explains that dislocations are not easily pinned with pillar size in the range of several microns. As the pillar size increases, dislocations take longer time to move across the specimen. Hence the retained dislocation density is much higher, which leads to a higher strain rate in pillar with a smaller size. Choi et al. (2013) attributes the higher strain rate to the increased contributions of free surface in the micro-pillar with a smaller diameter. In what follows, we manage to figure out the size dependent features observed in current study.

The derivation of the creep equation proceeds from the kinetic relation between the strain rate $\dot{\epsilon}$, the mobile dislocation density ρ_m and the average dislocation velocity v . Generally, the Orowan equation (Kassner, 2015) describes

$$\dot{\epsilon} = \rho_m v b \quad (12)$$

The evolution of mobile dislocations arises during the stress relaxation as a combinative processes caused by the multiplication and the annihilation of dislocations, i.e.

$$\dot{\rho}_m = \dot{\rho}_{mult} - \dot{\rho}_{anni} \quad (13)$$

where $\dot{\rho}_m$ denotes the rate of the density mobile dislocations, $\dot{\rho}_{mult}$ the rate of the density multiplication dislocations and $\dot{\rho}_{anni}$ the rate of density annihilation dislocations. The generation of dislocations can occur by emission of dislocations from a fixed number of sources, such as precipitate particles, grain boundaries, Frank-Read sources, and by multiplication of existing mobile dislocations (Caillard and Martin, 2003). According to Caillard and Martin (2003), whether the multiplication of dislocations arising or not during stress relaxation can be determined by performing transient tests before the yield stress. If a reloading yield stress following the relaxation test is smaller than that of tensile yield stress, this indicates that the multiplication of dislocations arises during the stress relaxation. To that end, multiple single relaxation tests were performed for the wire with a diameter of 35 μm . The first relaxation began with a starting stress of 25 MPa holding for 200 s. Then the sample was reloaded to a designed starting stress with an increment stress of about 20 MPa for the next relaxation. The corresponding stresses versus time results are shown in Fig. 11. One can see that the first reloading yield stress has a value of about 110 MPa (as shown in the insert). The reloading yield stress almost coincides with the monotonic tensile yield stress as described in Fig. 2. Thus we conclude that the multiplication of dislocations is not significant in our results. The annihilation of dislocations dominate the process during the relaxation transient. Here, two dislocation exhaustion

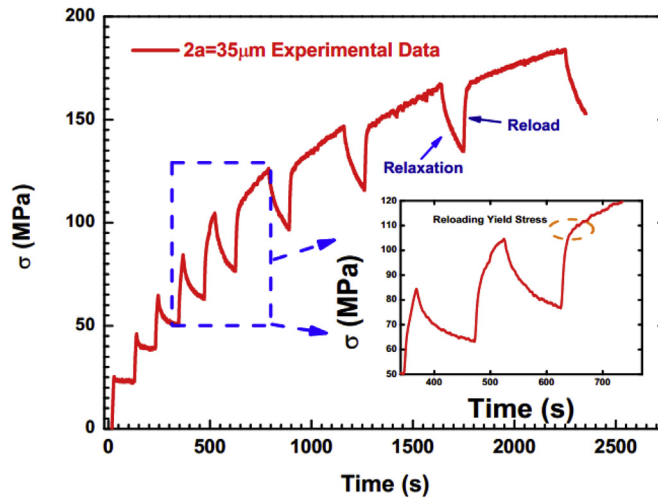


Fig. 11. Multiple single stress relaxation tests performed on the wire with a diameter of 35 μm at starting stress in elastic and plastic regime. Where a holding time of 200 s is kept in each relaxation transient.

mechanisms are mainly considered to estimate the size effect. They are the annihilation of mobile dislocations at grain boundaries in the grain of interior and that escaping from the free surface in the grain of near-surface zone.

According to Okamoto et al. (2014), Fritz et al. (2016), Gu et al. (2012) and Yang et al. (2012), it is found that the impacts of grains in the interior on the deformation are different with that near the surface. For instance, Fritz et al. (2016) concluded that dislocations can exit to the surface, stronger near-surface grains gain importance with decreasing sample dimension. This reduces dislocation pill-ups and local stresses, leading to the requirement of higher loads for plastic deformation. Gu et al. (2012) demonstrated that the deformation on the interior is bulk-like, while the surface allows more grain boundary sliding. Yang et al. (2012) demonstrated that in an inner grain, the source (normally the Frank-Read dislocation sources) produces more and more loops when pile-up at the grain boundary and/or interact with each other. Whereas in near-surface zone, typical double-ended Frank-Read sources can convert into single ended spiral sources that intersect and interact with free surface (Parthasarathy et al., 2007; Rao et al., 2007).

For the exhaustion of dislocations in grains of the near-surface zone, the hypothesis proposed by Yang et al. (2012) is employed. The whole specimen can be divided into two parts: a near-surface zone and a specimen of interior. The volume fraction of the near-surface zone f_s is approximated by

$$f_s = 4 \frac{\alpha}{r} \left(1 - \frac{\alpha}{r} \right) \quad (14)$$

in which α is the ratio the thickness of the near-surface zone s to grain size d , that is $\alpha = s/d$. And r is the ratio of wire diameter D to grain size d , i.e. D/d . It is found that f_s decreases with increasing r (Yang et al., 2012), namely the volume fraction of the near-surface zone is larger in sample with a smaller diameter. Relevant works illustrate that dislocations can escape easily from the surface in the grains of near-surface zone (Fritz et al., 2016; Parthasarathy et al., 2007; Yang et al., 2012). This results in a larger annihilation of dislocations as sample diameter decreases due to larger volume fraction of near-surface zone with constant grain size. In Wang et al. (2005), the high exhaustion rate of mobile dislocations was attributed to the ready availability of dislocation sources near the grain boundaries where dislocations can nucleate readily. But these grain boundaries dislocation sources are not regenerative, they can be used up quickly upon straining. Thus leads to a rapid decay of exhaustion of mobile dislocations. This exhaustion mechanism unlikely dominates here due to the multiplication of dislocations is not significant during the stress relaxation as concluded above. The higher exhaustion rate of mobile dislocation relates to the larger volume fraction of near-surface zone in the sample with smaller diameters. The mobile dislocations can easily escape from the free surface. As shown in Fig. 10, ρ_m/ρ_{m0} decrease to 0.52 for the wire with a diameter of 20 μm within $t = 20$ s.

For the annihilation of mobile dislocations at grain boundaries, a dislocation creep model is employed here. According to Webster (1966), the rate of multiplication of dislocations from a fixed number of sources N is proportional to νN , and the exiting mobile dislocations is proportional to $\nu \rho_m$. Therefore, the rate of multiplication dislocation density $\dot{\rho}_{mult}$ can be given by

$$\dot{\rho}_{mult} = \alpha_0 \nu N + \gamma \nu \rho_m \quad (15)$$

where α_0 and γ are the coefficients depending on stress and temperature.

The annihilation or immobilization of dislocations can occur by the interaction between mobile dislocations and a fixed number of obstacles M , and by the interaction between dislocations. The interaction between the mobile dislocations and the precipitate particles or grain boundaries results in a rate of dislocation annihilation or immobilization proportional to $\nu M \rho_m$. The rate that dislocations collide with each other is proportional to $\nu \rho_m^2$. The probability of a collision resulting in immobilization increases with

decreasing velocity. Thus, to a first approximation, the rate of the immobilization of dislocations by pair-wise interaction may be assumed to be independent of velocity and proportional to ρ_m^2 only. The total annihilation or immobilization rate $\dot{\rho}_{anni}$ can be expressed by:

$$\dot{\rho}_{anni} = \delta v M \rho_m + \lambda \rho_m^2 \quad (16)$$

where δ and λ are coefficients of proportionality which are functions of stress and temperature. They are constant in the stress relaxation test. Substituting Eqs. (16) and (15) into Eq. (13), we obtain the evolution rate of the mobile dislocation density

$$\dot{\rho}_m = \alpha_0 v N + (\gamma - \delta M) v \rho_m - \lambda \rho_m^2 \quad (17)$$

Under the condition of the quasi-steady strain rate during the stress relaxation, i.e. $\dot{\rho}_m = 0$ (Gu and Ngan, 2014; Tang et al., 2007). Eq. (17) can be written as

$$\alpha_0 v N + (\gamma - \delta M) v \rho_m - \lambda \rho_m^2 = 0 \quad (18)$$

The interaction between dislocations and grain boundaries is considered to be the predominant deformation mechanism during the stress relaxation, hence the magnitude of $\lambda \rho_m^2$ denoting dislocations collide with each other should be much less than that of $v M \rho_m$ representing the interaction between the mobile dislocations and obstacles. It is reasonable to assume that the term of $\lambda \rho_m^2$ can be neglected in Eq. (18). Subsequently, the density of mobile dislocations is derived as follows,

$$\rho_m \approx \frac{\alpha_0 N}{\delta M - \gamma} \quad (19)$$

Substituting Eq. (19) into the Orowan equation, one obtain the quasi-steady strain rate during the stress relaxation, i.e.

$$\dot{\epsilon} = \rho_m v b \approx \frac{\alpha_0 v N b}{\delta M - \gamma} \quad (20)$$

From Eq. (20), one can see that the quasi-static strain rate increases with increasing number of sources N , and with decreasing number of obstacles M . It has been generally accepted that grain boundaries can act as dislocation obstacles and sources (Brentnall and Rostoker, 1965; Kurzydowski et al., 1984; Shen et al., 1988; Worthington and Smith, 1964). Thus there exists a competition between mobile dislocations nucleated from grain boundary sources and that immobilized by grain boundaries. Under the grain size refinement condition, it is concluded that the contribution of multiplication of dislocations to mobile dislocation density is significant for FCC metals (Jiang and Zhang, 2015). And thus leads to an enhanced strain rate in sample with smaller grain size (Blum and Zeng, 2009; Duhamel et al., 2010; Rodriguez, 2004; Zhou et al., 2014). In our tests, wires with different diameters possess similar grain sizes. The number of grain boundaries increases with increasing the wire diameter. This results in increasing numbers of nucleation sources N and obstacles M . As discussed above, the multiplication of dislocations is not significant during the process here. Grain boundaries mainly act as the obstacles to impede dislocation motivation rather than as dislocation sources here. To verify this, TEM images of deformed wire with a diameter of 35 μm after repeated relaxation tests with a starting stress of 120 MPa are shown in Fig. 12 (a)-(b). The TEM samples were prepared by cutting sheets in a Quanta 3D Dual Beam FIB system both in the cross-section and longitudinal-section. And TEM images were recorded in a TEM (Tecnai G2 20). From Fig. 12(a) and (b), one can see that grain

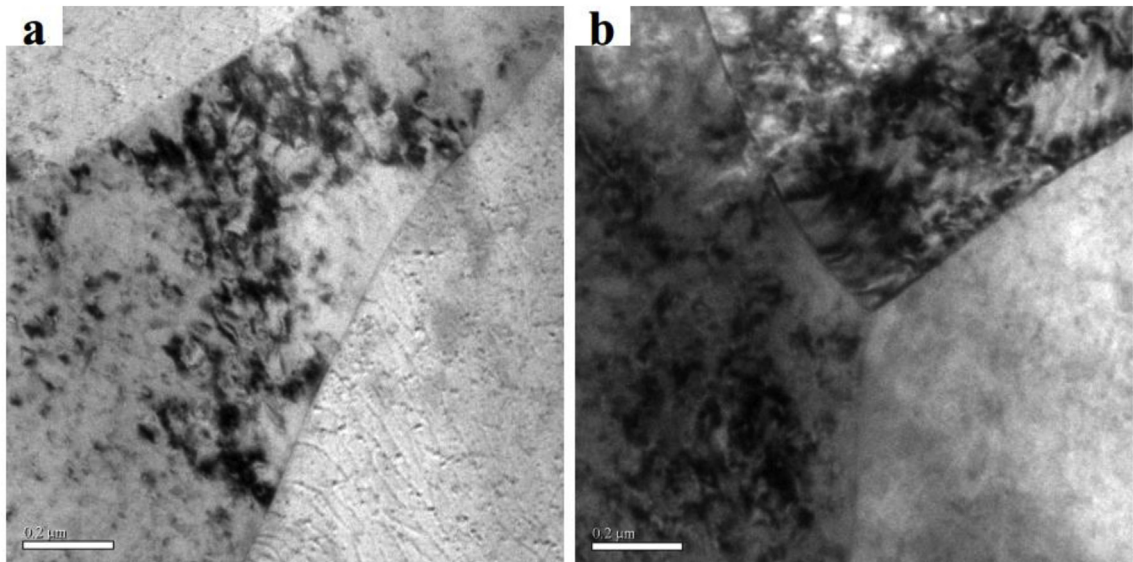


Fig. 12. TEM images for deformed wire with a diameter of 35 μm . (a) Prepared in the cross-section, (b) in the longitudinal-section.

boundaries impede dislocation glide from one grain to neighboring one. The contribution of number of grain boundary dislocation sources N is less important than that of dislocation obstacles M in Eq. (20). Thus smaller mobile dislocation density occurs in samples with larger number of grain boundaries. Besides, more obstacles will reduce the possibility of localized slip on a given slip trace and lower the average dislocation velocity (Norfleet et al., 2008), this leads to a smaller strain rate in sample with larger number of obstacles.

We propose that the bilinear relation occurs as a result of a transition of the dominate annihilation dislocations mechanism in different regime. For the wire with the smallest diameter, the annihilation of dislocations in the primary of the relaxation regime is dominated by the dislocation escaping from the free surface due to the large volume fraction of near-surface zone. In this range, mobile dislocations can easily escape from the surface. This leads to a rapid drop of dislocation evolutions. After the primary transient regime, few newly dislocations can be nucleated to compensate the mobile dislocation exhausted from the surface in the primary relaxation regime. This arises on the basis of that the multiplication of dislocations is not significant during the relaxation transient. And grain boundaries act as obstacles to prevent dislocations in the grain of interior to gliding into that of near-surface zone. Hence, the annihilation of dislocations mainly attribute to the dislocations immobilized at grain boundaries in this range. While in the wires with larger diameters, dislocations escaping from the surface is less significant than that annihilated at grain boundaries even in the primary relaxation transient due to the large volume fraction of the interior grains. The annihilation of dislocations is dominated by the dislocation immobilized at the grain boundaries in the total relaxation transient for sample with large volume fraction of the interior.

In Gu and Ngan (2014), dislocations are considered not to be easily pinned by the obstacles, i.e., precipitates, at a few microns. A decrease of strain rate in a wire with a smaller diameter is also observed by Gu and Ngan (2014). But they pointed out that the trend would be inverse in the tens of microns, where dislocations may experience a higher chance of being pinned and immobilized by the obstacles. In our tests, the diameter of the wires are in the range from 20 to 100 μm . Remarkably, our results are in agreement with the expectations of Gu and Ngan (2014). Hence, when the propagation of dislocations encounters grain boundaries in samples with diameters in tens of micron meters, the dislocation have more possibility to be pinned/hindered by the grain boundary than to move across them.

4.3. Size effect in activation volume

For the grain size dependent characteristic in the value of activation volume, an analogous Hall-Petch relation between the activation volume parameter $V(d)$ and the grain size d has been proposed as (Armstrong and Balasubramanian, 2017; Armstrong and Rodriguez, 2006; Rodriguez, 2004):

$$\frac{1}{V(d)} = \frac{1}{V_0} + \frac{k_\epsilon}{2m_T\tau_c V_c} d^{-1/2} \quad (21)$$

where V_0 is the activation volume associated with the intra-grain dislocation mechanism, i.e. intersection of lattice dislocations (Armstrong and Balasubramanian, 2017). And V_c denotes the activation volume relevant to the grain boundary mediated mechanisms, m_T the Taylor factor ($m_T \approx 3.1$). τ_c can be determined from the classical Hall-Petch model of dislocation pile-up. In the twin thickness-dependent of time-dependent plastic deformation, Lu et al. (2009) considered the influence when impinging dislocations react with twin boundaries. An additional term, i.e. $\lambda^{-1/2}k_{TB}/2m_T\tau_c V_{TB}$, is involved to include twin boundary effect. In which V_{TB} is the activation volume associated with the twin boundary mediated mechanisms, and λ the twin thickness.

By considering the different contributions of the grain in interior and near-surface zone to time-dependent deformation, we hypothesize that the activation volume V is divided into two parts: V_i and V_s . Where V_i denotes the contribution of grains in the interior and V_s represents that of the near-surface zone. Following Yang et al. (2012), the activation volume V can be rewritten as

$$V = f_s V_s + (1 - f_s) V_i \quad (22)$$

where f_s is volume fraction of the near-surface zone, defined as Eq. (14). Normalization Eq. (21) by V_i and combination the setting ratio of $\xi = V_s/V_i$ with $0 < \xi \leq 1$, one can obtain that

$$\frac{V}{V_i} = f_s \xi + (1 - f_s) \quad (23)$$

Substituting Eq. (14) into Eq. (22) leads to

$$\frac{V}{V_i} = 1 - 4(1 - \xi) \frac{\alpha}{r} \left(1 - \frac{\alpha}{r}\right) \quad (24)$$

Here, two extreme cases should be discriminated as that in Yang et al. (2012). One case is for very large ratio r , i.e. the wire contains numerous grains in the sample cross-section. This leads to $f_s = 0$, $\xi = 1$ and $V = V_i = V(d)$, where $V(d)$ is that defined in Eq. (20). The second is for $r < 2$, the validity if Eq. (23) is held when $\alpha = s/d < r/2$. Based on Eq. (23), the values of activation volume with respect to the ratio r and $d^{-1/2}$ are shown in Fig. 13(a) and Fig. 13(b), respectively. In which V_i is calculated with $V_i = V(d)$. And the parameters in Eq. (20) are assumed as: $V_0 = 148 \text{ b}^3$, $k_\epsilon = 5.0 \text{ MPa mm}^{-1/2}$ (Lu et al., 2009a), and $\tau_c V_c = 3.1 \times 10^{-20} \text{ J}$ (Rodriguez and Armstrong, 2006). From Fig. 13(a), one can see that the activation volume increases evidently with increasing r when r is less than a critical value of about 25. And the activation volume increase with the increasing ξ , while it decreases with the increasing α . Within large r , the activation volume approaches to a constant value. Fig. 13(b) describes that the activation volume decreases with $d^{-1/2}$.

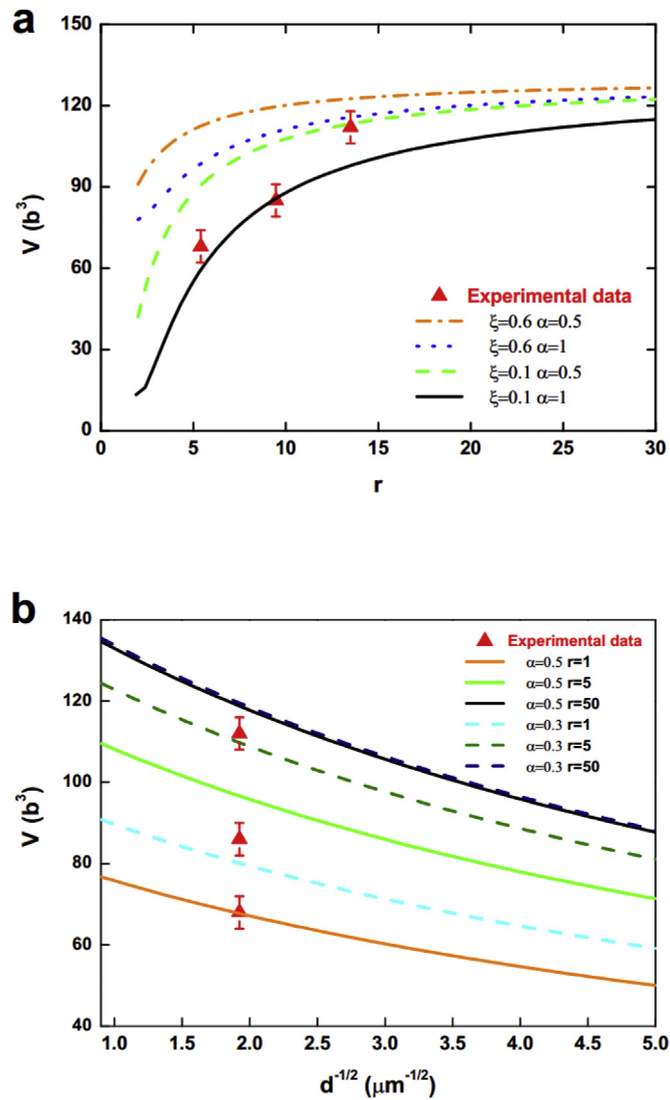


Fig. 13. The value of activation volume V (a) with respect to the ratio of the wire diameter to grain size r , (b) versus $d^{-1/2}$ at different ratio r values. Both the modeled and experimental results are present, the value of activation volume is selected with a starting stress of 120 MPa. And in (b) the parameter ξ is chosen as 0.6 (Rao et al., 2007; Yang et al., 2012).

Table 3

The value of activation volume V versus grain size (or twin thickness*) d and sample size t measured from various tests at room temperature.

Method	Materials	d	D	V	Ref.
Stress Relaxation	Polycrystalline Cu wire	3.7 μm	20 μm	47.6 b^3	Present
		3.7 μm	35 μm	61.4 b^3	
		3.7 μm	50 μm	88.1 b^3	
Creep	nc Ni	30 nm	0.6 μm	1.15 b^3	Choi et al. (2013)
		30 nm	1 μm	1.30 b^3	
		30 nm	2 μm	2.18 b^3	
Stress Relaxation	ng Cu	70 nm	300 μm	208.9 b^3	Yang et al. (2016b)
		50 nm	300 μm	49.7 b^3	
Stress Relaxation	ufg Cu	0.4–0.6 μm	20 μm	350 b^3	Lu et al. (2009)
	nt-35 Cu	35 nm*	20 μm	36 b^3	
	nt-15 Cu	15 nm*	20 μm	20.7 b^3	

This agrees with that observed in the works (Armstrong and Balasubramanian, 2017; Armstrong and Rodriguez, 2006; Rodriguez, 2004). The activation volume in Eq. (23) can reduce to that predicted in Eq. (20) with large r , e.g. $r = 50$. The values of activation volume versus grain size or twin thickness and specimen size measured from various tests are illustrated in Table 3. In Yang et al. (2016b) and Lu et al. (2009), samples with different grain size or twin thickness possess same specimen size. An elevated value of activation volume is observed in a sample with larger grain size or twin thickness. The observations accord with the prediction in Eq. (23). In Choi et al. (2013), samples with different diameter possess same grain size of about 30 nm. A larger value of activation volume is observed in a sample with larger diameter. For instance, the value of activation volume is 1.15 b^3 , 1.30 b^3 , 2.18 b^3 for pillar with a diameter of 0.6 μm , 1 μm , 2 μm , respectively. This is consistent with our findings. And the phenomenon also can be predicted by Eq. (23). That is the value of activation volume increases with increasing the ratio of the wire diameter to grain size. To sum up, the size dependence in the time-dependent plasticity can be affected by the grain size incorporating with the specimen size.

4.4. Stress dependence in activation parameters

Stress dependences in activation parameters are observed here. One can see that the larger the starting stress, the rapider the stress relaxation. The activation volume decreases with the increasing applied starting stress, while the strain rate increases with that. In what follows, we will briefly demonstrate the process causing the stress dependent features observed in our tests. A thermally activated process can be generally expressed by the Arrhenius relation within a certain temperature range. The plastic strain rate can take the general form as (Mukherjee et al., 1968):

$$\frac{d\varepsilon_p(\sigma, T)}{dt} = \frac{d\varepsilon_0}{dt} \left(\frac{\sigma}{\sigma_0} \right)^n \exp \left(-\frac{\Delta G(\sigma)}{kT} \right) \quad (25)$$

where ε_0 is a stress-independent parameter, σ the applied stress, σ_0 a reference stress, n the stress exponent, and ΔG the apparent activation energy, respectively. A common and simple relationship between the apparent activation energy and stress is expressed by (Chang et al., 1999)

$$\Delta G(\sigma) = \Delta G_0 - \frac{\sigma}{\sqrt{3}} V \quad (26)$$

where ΔG_0 is a stress-independent activation energy, and V the nominal activation volume conjugated to the shear stress. Substituting Eq. (25) into Eq. (24), one can obtain

$$\frac{d\varepsilon_p(\sigma, T)}{dt} = \frac{d\varepsilon_0}{dt} \left(\frac{\sigma}{\sigma_0} \right)^n \exp \left(-\frac{\Delta G_0}{kT} + \frac{V\sigma}{\sqrt{3}kT} \right) \quad (27)$$

According to Yang et al. (2016a), the stress exponent n has the relation with strain rate sensitivity m as follows, that is

$$n = \frac{V\sigma}{\sqrt{3}kT} - \frac{1}{m} \quad (28)$$

Taking the sample with a diameter of 50 μm as an example, the calculated stress exponent n has a value of 5.96, 4.94, 1.32 with applied stress of 92 MPa, 77 MPa, 62 MPa, respectively. In Eq. (26), a positive value of stress exponent will lead to an elevated strain rate as the applied stress increases. This agrees with our results (see Fig. 8), also in that of Choi et al. (2013). As for the relation between the activation volume and the applied stress, our results meet the principle of Cottrell-Stokes law, namely $V \propto 1/\sigma$ (Bochniak, 1993; Duhamel et al., 2010). The value of activation volume can be estimated in terms of applied stress as (Armstrong and Balasubramanian, 2017; Choi et al., 2013):

$$V = \sqrt{3}kT \left(\frac{\partial \ln \dot{\varepsilon}}{\partial \sigma} \right) = \frac{\sqrt{3}kT}{m\sigma} \quad (29)$$

This coincides with our findings and that observed in Oksiuta et al. (2011), Kim et al. (2012b) and Kubát and Rigdahl (1976). The elevated apparent activation volumes at small strains can be visualized as an artifact induced by the evolving microstructure (Mohanty et al., 2016). At low strains, the deformation microstructure (e.g., dislocation content, grain boundaries, etc.) is limited and evolves rapidly during the relaxation transient used to measure the apparent activation volume. At higher strains, microstructural evolution is relatively not as substantial over the same period of time.

5. Conclusions

The tensile stress relaxation behavior of thin copper wires with diameters 20, 35, 50 μm is experimentally investigated at room temperature in this study. It is revealed that the tensile stress relaxation does occur at low stress below the yield strength. Significant size and stress dependences are clearly observed. That is at a given stress, a rapider stress relaxation accrues in a sample with smaller diameter. And at a given diameter, the stress relaxation is accelerated with increasing the starting stress. Moreover, size dependent activation parameters are also observed. The values of activation volume decrease with decreasing the sample diameter. While the strain rate and strain rate sensitivity parameter increase with decreasing the sample diameter. A bilinear relation between the logarithmic strain rate and the logarithmic stress is observed only for sample with a diameter of 20 μm . The intersection of dislocations with grain boundaries is considered to dominate the deformation mechanism during the stress relaxation.

A rapid exhaustion of mobile dislocations is observed in the wire with smaller diameter in the primary of relaxation. It is attributed to annihilation of dislocations easily escaping from the surface due to large fraction volume of the near-surface zone. After the primary regime, the higher possibility of dislocations being pinned or immobilized by grain boundaries in a larger-diameter sample is accounted for the observed size dependence. The dominated exhaustion mechanism during the relaxation regime is discussed. We propose that in the sample with smallest diameters, the exhaustion of dislocations from the surface dominates the process in the primary of relaxation. After that, it is the dislocations annihilated at grain boundaries that dominates. The transition is accounted for the observed bilinear relation. For wire with larger diameters, the dislocations annihilated at grain boundaries dominates process in the total relaxation regime due to a large volume fraction of interior. Here, a model is proposed to predict size effect in the activation volume. It indicates that the activation volume increases with increasing the ratio of diameter to grain size $r = D/d$ with r below a critical value. And the activation volume increases with increasing the grain size. To further investigate the influence of r on the time-dependent plasticity, stress relaxation tests at various grain sizes are needed for samples with different diameters.

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